

Software Project Management

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Before You Start: Assessing Value and Risks

- Projects create new products, new services, or new capabilities. The first step of a sound management process is to understand whether the new products, services, or capabilities are worth our effort.
- A project's importance usually depends on two main factors:
 - The **value** generated by the project
 - The **risks** associated with the project

Project Value: Aspects to Consider

- Three main factors determine the value generated by a project:
 - **Direct and indirect value.** The value of a project is not just about the money it earns directly or through its results. Other things can also make a project more or less valuable. These include its social and environmental effects, how it improves or harms the organization's image, the publicity it creates, the chance to enter a new market, and the knowledge or skills gained from it. All of these can add to or reduce the value of a project.
 - **Sustainability.** Many IT projects start without a clear plan for how their results will continue after the project ends. As a result, the outcomes may not last long once the project and its resources are finished. It is important to think about the ongoing costs to keep the project's results running and how they will survive after the project is over. For example, if the project creates an app, there should be a plan for updates and maintenance. This helps decide whether the project is worth doing in the first place.

- **Alignment with the strategic objectives of the organization.** Before starting a project, it is important to make sure it matches the organization's main goals. A project that supports these goals is more likely to be given higher priority. If the project does not align with the strategy, it might not be a good use of time, money, and effort. Ensuring alignment helps make sure the project's results will benefit the organization in the right way.

Project Risks: Aspects to Consider

- Various factors determine the risk profile of a project. Among them are:
 - **Resource availability.** Projects need resources—such as people, money, and technology —at the right time. If these resources are not available when needed, the project may face delays or even fail.
 - **Timing.** Many projects have a limited time window in which their results must be delivered. If the delivery happens too early or too late, the results might be useless. For example, building a rocket to travel to another planet requires launching at a specific time to take advantage of the planets' positions (known as the “slingshot” effect). Missing this window could make the mission impossible or ineffective.
 - **Technical difficulty or uncertainty.** Some projects depend on solving difficult technical problems. If the team cannot find a solution, the project may not succeed. Uncertainty in technology or methods increases the risk.

- **Project environment and constraints.** Projects are affected by limitations from inside and outside the organization. Internal and external stakeholders may try to influence the project in positive or negative ways. Also, rules, regulations, and standards can restrict what the project is allowed to do. These constraints can increase the risk of failure.

Techniques to Assess Value and Risks:

Financial Methods

- **Payback:** The payback method is the simplest way to check a project's financial value. It measures how long it will take for the project to earn back the money that was invested in it. To use this method, we estimate the project's costs and incomes for each year and then calculate the profits or losses at the end of each year. The "payback" is the year when the total earnings cover all the expenses and the project starts making a profit. A shorter payback period is better because the investment is recovered faster.
- **Return of Investment:** The ROI method is used to fix some of the weaknesses of the payback method. ROI measures how much profit we earn for every dollar invested. It is calculated from the annual profit, which is the average profit made per year. To find the annual profit, we divide the total profit by the project's duration in years. A higher ROI means the project gives better returns for the money invested.

$$\text{Annual profit} = \frac{\text{incomes} - \text{expenses}}{\text{project duration}}$$

The ROI is then computed by dividing the *annual profit* by the total project expenses:

$$\text{ROI} = \frac{\text{annual profit}}{\text{expenses}}$$

- **Net Present Value and Internal Rate of Return:** Payback and ROI have a limitation—they do not consider inflation, which means that money in the future will be worth less than the same amount of money today.
- The net present value method (often called NPV) solves this problem by including the effect of inflation in its calculations. If we can make a reliable estimate of the inflation rate, we can adjust the value of the project's future costs and incomes so they reflect their real value in today's terms.
- In particular, when using NPR, profit and losses are computed using the following formula:

$$\text{Value} = \frac{1}{(1 + r)^i} * \text{amount}$$

- Where r is the inflation rate, amount is the net profit at year i , and value is the current value of amount. Note that the first project year is year 0.

Applying Financial Methods: An Example

	<i>Project A</i>	<i>Project B</i>
Year 0	–€20,000.00	–€30,000.00
Year 1	–€10,000.00	–€30,000.00
Year 2	€40,000.00	€50,000.00
Year 3		€100,000.00

- If we use the payback method, we select Project A, since it has a shorter payback period. In fact, year 0 is forecast to end with losses, for Project A. So will year 1. At the end of year 2, however, the financial statement will show earnings of EUR 10, 000.

Project B's payback is 4 years. Years 0 and 1 end with losses. In year 2, project B earns profits of €50,000, but these are not yet sufficient to cover the expenses of the first 2 years. In year 3, however, the project pays back the initial investment.

If we use the ROI method, we select Project B, since it has the highest ROI.

Project A, in fact, has an ROI of 11%, computed as follows:

$$\text{Annual profit} = \frac{\text{€}40,000 - (\text{€}20,000 + \text{€}10,000)}{3} = \text{€}3333$$

$$\text{ROI} = \frac{\text{€}3333}{\text{€}300,000} = 11\%$$

Score Matrices

- Financial methods help determine the financial viability of a project, but tell nothing about the project characteristics that are not measurable with profits and losses.
- Therefore, other methods have been proposed to assess a project.
- One of the simplest is the **score matrix**, which allows one to measure a project along several dimensions and assign it a value.
- A score matrix is a list of project criteria, each of which is assigned a weight, which measures the importance the criteria for the organization we work for.

- When we evaluate a project using a score matrix, we measure how well the project satisfies each criterion, by assigning a number from 1 (very low) to 5 (very high).
- We then multiply the scores with the weights and sum all values. Projects scoring a higher value are more desirable than projects with a lower score.
- Projects can also be compared side by side; hence, the use of the term “matrix” in the name of the technique.

<i>Factor</i>	<i>Description</i>	<i>Weight</i>	<i>Project 1</i>		<i>Project 2</i>		<i>Project 3</i>	
			<i>Value</i>	<i>Total</i>	<i>Value</i>	<i>Total</i>	<i>Value</i>	<i>Total</i>
Profit >30%	The project will yield a profit >30% if no exceptional events occur	3	4	12	4	12	4	12
Low-risk profile	The project does not present particular risks. That is, there is no risk with a very high impact	2	2	4	3	6	3	6
Schedule is not tight	Project delivery does not require activities to be performed in a very tight schedule	3	3	9	2	6	2	6
Manageable complexity	The complexity is manageable	2	1	2	1	2	1	2
Consistent with current business	The project is mainstream with the activities of the organization	1	1	1	1	1	1	1
Stakeholders	Stakeholders are difficult to manage	−4	2	−8	4	−16	3	−12

Stakeholder Analysis

- Stakeholders can exert quite a lot of influence on a project and determine the success or failure of a project.
- Understanding how stakeholders can influence (positively or negatively) a project, is good practice to assess the project's chances of success and to define a stakeholder management policy.
- Various ways have been proposed to classify stakeholders. In general, stakeholders are classified in two dimensions:
 1. The **power** they can exert in the project
 2. The **interest** they have in the project

Assessing Sustainability

- Evaluating the operational cost of a project outputs helps assess the long-term benefits of a project and what additional actions a project should include to ensure that its results will live after the project ends.
- This analysis is relevant in several high risk or highly constrained projects, such as projects to start new companies, research projects, and projects to foster economic development.
- Often, in fact, these projects deliver benefits as long as steady financing is available; little or nothing survives when the project ends.

The Project Feasibility Document

- The project feasibility document closes the assessment phase by describing the main characteristics of a project and by analyzing, using techniques presented earlier, the value and risks of a project.
- The document can be used to formally authorize the start of a project.
- Thus, a feasibility document contains at least the following information:
 - A **statement of work**, which describes the scope of the project
 - The **business objectives** of the project -> impact of its outputs
 - A summary of the **project budget**, which forecasts expenses and incomes

- A summary of the project **milestones**, which is a rough schedule of the project identifying the most important events
- An analysis of the **stakeholders**
- The project **risks**
- Possible **alternatives** to the project, such as a make-or-buy decision
- An **evaluation** of the project and alternatives, using the techniques described above

Formalizing process...

- After assessing Value and Risks, one of the first activities is fixing and having stakeholders agree on the project scope.
- This, in fact, constitutes the basis for any further management activity, namely, the definition of the work to be performed, schedule and budget, and the skills required by the team that will perform the work.
- Project goals, together with the schedule and the budget, are also the basis for contractual agreements with the clients.

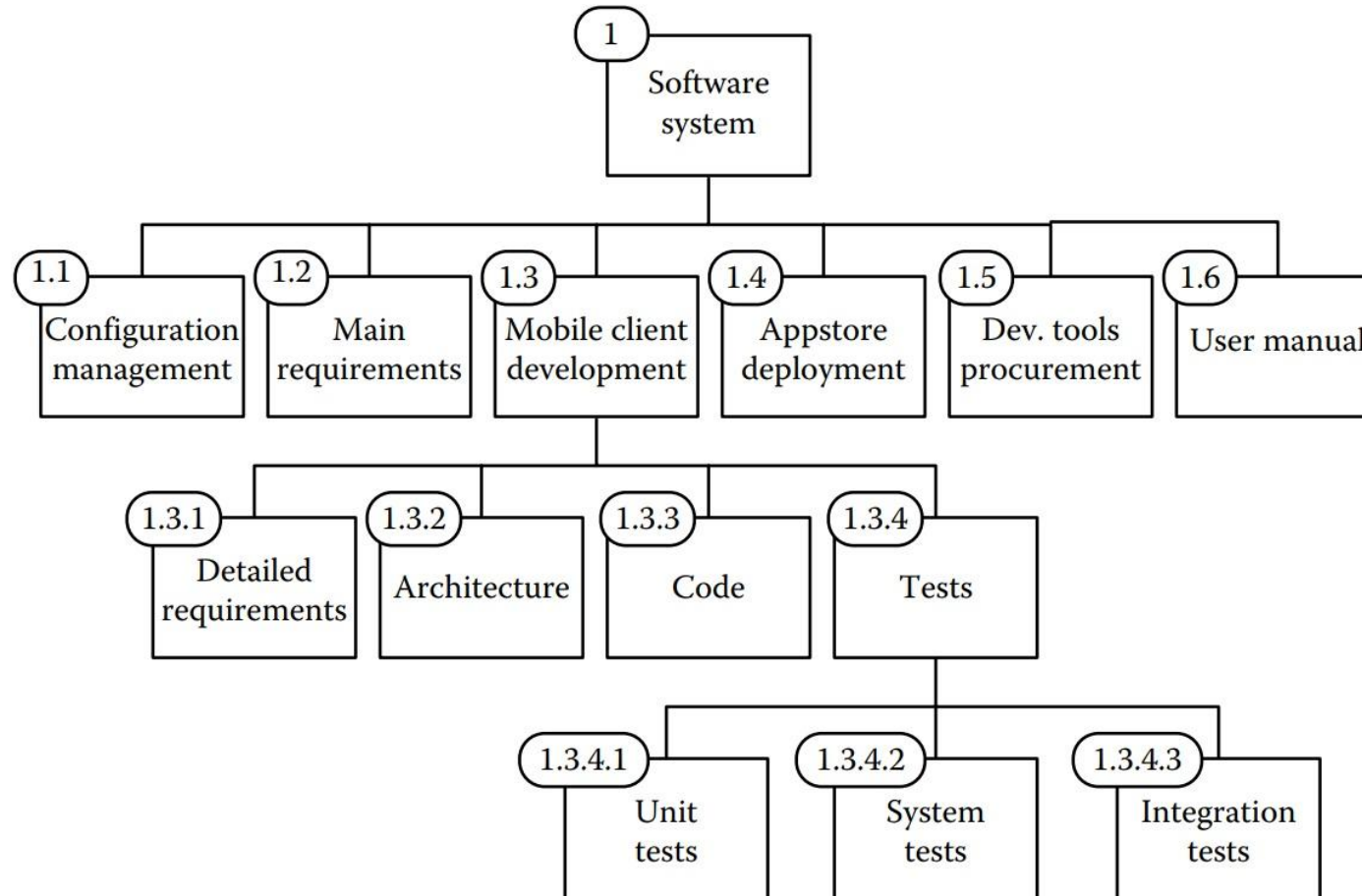
Deciding the Work

- Now that we have properly described the main goals and the boundaries of our project; we can start identifying the activities that we need to carry out in the project.
- This is very often accomplished with a **work breakdown structure** or WBS from now on.
- The notation, developed in the 1960s alongside the program evaluation and review technique (PERT), is defined in Project Management Institute (2004) as “a (deliverable-oriented) hierarchical decomposition of the work to be executed by the project team to accomplish projects objectives and create the required deliverable.”

- Today, the technique is widely adopted and many project management and process standards make us require a WBS to be built for each major program or project and suggest its use in any project, big or small, when it can be practically done so.
- A WBS establishes the basis for:
 - Defining the work to be performed in a project
 - Showing how various activities are related to the project objectives
 - Establishing a framework for defining, assigning, and monitoring work and costs
 - Identifying the organizational elements responsible for accomplishing the work.

Building a WBS

- A WBS is a tree in which the root node represents a project or its main output. Each level of decomposition shows how a node of the tree can be structured and organized in more elementary work components.
- WBSs come in two different notations. The first is graphical: the WBS is shown as a tree unfolding from top to bottom. The second is textual and similar to the table of contents of a book.



- The Project Feasibility Document -> Deciding the Work -> WBS

WBS Decomposition Styles

- Different decomposition styles allow one to build a WBS.
- In a **product-oriented** WBS, the decomposition proceeds by identifying the items that must be developed to build deliverables.
- In a **process-oriented** WBS, the decomposition proceeds by taking into account the activities that are necessary to carry out the project. One advantage of the process-oriented decomposition is that it can include activities, such as management, which are not directly related to the development of a product, but which are still necessary in a project.
- To get the best of both worlds, one can use a hybrid WBS. A hybrid WBS contains both process- and product-oriented nodes.

WBS Dictionary

- A WBS dictionary helps to annotate each element of the WBS with more detailed and structured information, such as
- Title and item number, to connect the description to an element of the WBS
- Detailed description of the element, including, for instance, quantities, relevant associated work, and contractual items, where applicable...

WBS Construction Methodologies

- WBSs can be built **top-down**, **bottom-up**, or by **analogy**.
- In the **top-down** approach, the construction of the WBS proceeds from the top level down to the leaves. It is usually best suited for projects that are well known or whose structure is clear.
- In the **bottom-up** approach, the process proceeds in the opposite way. First, the leaves of the WBS are identified and then these are grouped in homogeneous items, thus giving structure to the WBS. It is best suited for projects that are new for a company.
- The third option, construction by **analogy**, starts from an existing WBS, which is adapted and customized for the project at hand. It can be very effective when an organization standardizes the structure of its projects.

